

Slope & Linear Relationships

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Slope = rise $\div$ run. In $y = mx + b$, m is the slope and b is the y-intercept. A steeper line has a larger slope. Slope can be positive (up), negative (down), zero (flat), or undefined (vertical).

Round 1 — Retrieve It

1. Write the formula for slope in words.

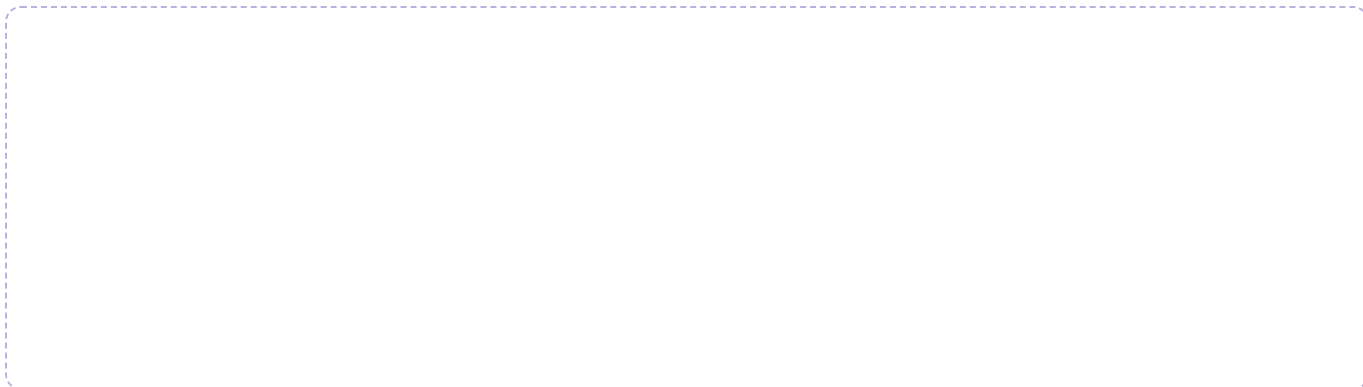
2. In $y = mx + b$, what do m and b each represent?

3. Which has a larger slope: a steep line or a nearly flat line?

4. Name the four kinds of slope.

Round 2 — Sketch It

On a coordinate grid, draw a line that rises 2 units for every 3 units it runs to the right. Mark the rise and the run, and label the slope as a fraction.



Slope & Linear Relationships — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A line rises 6 and runs 2. What is its slope?

2. For $y = 4x + 3$, name the slope and the y-intercept.

3. Find the slope between the points (1, 2) and (4, 8).

Round 4 — Reason from Evidence

Line A is $y = 2x + 1$ and Line B is $y = 5x + 1$. Use the slopes to decide which line is steeper, explain how you know, and describe what b tells you about where each line crosses the y-axis.

Slope & Linear Relationships — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain how the slope of a line tells you how steep it is and which way it tilts. Use one example that is not printed on the cue card.

C**Connect**

Connect slope to a real rate, such as dollars earned per hour. In $y = mx + b$, what would m and b mean in that situation?

E**Extend**

Create your own equation in the form $y = mx + b$. State its slope and y-intercept, then describe the line as positive, negative, zero, or undefined slope.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Slope & Linear Relationships

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Slope = rise $\div$ run (the change in y over the change in x).
2. m is the slope; b is the y-intercept (where the line crosses the y-axis).
3. A steep line has the larger slope.
4. Positive, negative, zero, and undefined.

Sketch It — look for

A line with rise 2 and run 3 marked on the grid, labeled slope = $\frac{2}{3}$.

Apply It

1. slope = rise $\div$ run = $6 \div 2 = 3$.
2. slope **m = 4**; y-intercept **b = 3**.
3. slope = $(8 - 2) \div (4 - 1) = 6 \div 3 = 2$.

Reason from Evidence — look for

Line B has slope 5 and Line A has slope 2, so **Line B is steeper** because a larger slope means a steeper line. Both have $b = 1$, so both cross the y-axis at (0, 1).

ACE — Extend look-for

Many answers possible. Check that the equation is in $y = mx + b$ form and the slope and y-intercept are correctly identified and described.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.